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PACKAGE AND CLOSURE FOR PACKAGE

Abstract

A package includes a container and a closure. The container includes a container brim, a container floor spaced apart from the container brim, and a container sidewall extending between and interconnecting the container brim and the container floor. The closure is configured to couple to the container brim to close selectively an open mouth of the container defined by the container brim.

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Background/Summary

PRIORITY CLAIM [0001] This application claims priority under 35 U.S.C. § 119(e) to U.S. Provisional Patent Application No. 63/562,940, filed Mar. 8, 2024, which is expressly incorporated by reference herein in its entirety.

BACKGROUND

[0002] The present disclosure relates to a package, and particularly to a package including a container and a closure. More particularly, the present disclosure relates to a closure including one or more polymeric materials.

SUMMARY

[0003] According to the present disclosure, a package includes a container and a closure. The container includes a container brim, a container floor spaced apart from the container brim, and a container sidewall extending between and interconnecting the container brim and the container floor. The closure is configured to couple to the container brim to close selectively an open mouth of the container defined by the container brim.

[0004] In illustrative embodiments, the closure includes a closure top wall, a closure sidewall, and a container-sealing flange. The closure side wall is coupled to the closure top wall and extends downwardly from the closure top wall to engage the container to mount the closure to the container. The container-sealing flange is coupled to an upper end of the closure sidewall and is configured to mount with the container brim of the container and close the open mouth.

[0005] In illustrative embodiments, wherein the container-sealing flange includes a support layer coupled to the closure sidewall and a seal ring bonded to a lower surface of the support layer. The seal ring is configured to establish: (i) a sealed, unopened state, (ii) a first-use opened state, and (iii) an unsealed, closed state of the package. In the sealed, unopened state, the seal layer is sealed with an upper surface of the container brim to block effluents from entering the container interior. In the first-use opened state, the closure is separated from the container after being opened for the first time. In the unsealed, closed state, the closure is reinstalled on the container. The seal layer remains attached to the closure after being opened for the first time so that the closure and the seal layer are not discarded separately from one another and can be recycled with one another.

[0006] In some embodiments, the seal layer is ultra-sonically welded to the container brim. In some embodiments the seal layer is heated by a heating device to bond the seal layer to the container brim. In some embodiments, the seal layer is sonically welded to the container brim. In each case, the seal layer provides a heat seal between the support layer and the container brim around the entire circumference or periphery of the container brim.

[0007] Additional features of the present disclosure will become apparent to those skilled in the art upon consideration of illustrative embodiments exemplifying the best mode of carrying out the disclosure as presently perceived.

Description

BRIEF DESCRIPTIONS OF THE DRAWINGS

[0008] The detailed description particularly refers to the accompanying figures in which:

[0009] FIG. **1** is a perspective view of a package including a container and a closure configured to be coupled to the container to seal and block removal of product from the container;

[0010] FIG. **2** is a cross section of the package taken along line **2-2** in FIG. **1** showing the closure including a top wall, a sidewall extending downwardly from the top wall, and a container-sealing flange configured to engage with a brim of the container and establish a seal with the container;

[0011] FIG. **3** is an enlarged view of a portion of FIG. **2** showing the container-sealing flange including a support layer formed integrally with the rest of the closure and a seal layer bonded to a lower surface of the support layer and including a peelable material to allow release of the closure from the container;

[0012] FIG. **4** is a top perspective view of the closure showing the closure sidewall including a plurality of reinforcement ribs spaced circumferentially about a central axis of the closure and configured to reinforce the closure sidewall during installation and removal of the closure from the container;

[0013] FIG. **5** is a cross section of a portion of the closure taken along line **5-5** in FIG. **4** showing the closure sidewall further including an annular retainer ring located directly adjacent to the container-sealing flange and configured to engage with the container to retain the closure on the container following opening of the package for the first time and separating the seal layer from the container;

[0014] FIG. **6** is an enlarged view of a portion of FIG. **5** showing the support layer and the seal layer making up the container-sealing flange;

[0015] FIG. **7** is a flow chart showing a forming and assembly process for the package of FIG. **1**;

[0016] FIG. **8** is a perspective view of a second embodiment of a package including a container and a closure configured to be coupled to the container to seal and block removal of product from the container;

[0017] FIG. **9** is a cross section of the package taken along line **9-9** in FIG. **8** showing the closure including a top wall, a sidewall extending downwardly from the top wall, and a container-sealing flange configured to engage with a brim of the container and establish a seal with the container;

[0018] FIG. **10** is an enlarged view of a portion of FIG. **9** showing the container-sealing flange including a support layer formed integrally with the rest of the closure and a seal layer bonded to a lower surface of the support layer and including a peelable material to allow release of the closure from the container;

[0019] FIG. **11** is a top perspective view of the closure showing the closure sidewall including a plurality of reinforcement ribs spaced circumferentially about a central axis of the closure and configured to reinforce the closure sidewall during installation and removal of the closure from the container;

[0020] FIG. **12** is a cross section of a portion of the closure taken along line **12-12** in FIG. **11** showing the closure sidewall further including an annular retainer ring located directly adjacent to the container-sealing flange and configured to engage with the container to retain the closure on the container following opening of the package for the first time and separating the seal layer from the container;

[0021] FIG. **13** is top plan view of the closure of FIG. **1**;

[0022] FIG. **14** is a right side view of the closure of FIG. **1**;

[0023] FIG. **15** is a rear view of the closure of FIG. **1**;

[0024] FIG. **16** is a left side view of the closure of FIG. **1**;

[0025] FIG. **17** is a front view of the closure of FIG. **1**;

[0026] FIG. **18** is a bottom view of the closure of FIG. **1**;

[0027] FIG. **19** is a bottom perspective view of the closure of FIG. **1**;

[0028] FIG. **20** is top plan view of the closure of FIG. **7**;

[0029] FIG. **21** is a right side view of the closure of FIG. **7**;

[0030] FIG. **22** is a rear view of the closure of FIG. **7**;

[0031] FIG. **23** is a left side view of the closure of FIG. **7**;

[0032] FIG. **24** is a front view of the closure of FIG. **7**;

[0033] FIG. **25** is a bottom view of the closure of FIG. **7**; and

[0034] FIG. **26** is a bottom perspective view of the closure of FIG. **7**.

DETAILED DESCRIPTION

[0035] A package **10** in accordance with the present disclosure includes a container **12** and a closure **14** as shown in FIG. **1**. The container **12** is formed to include a container interior **16** configured to store a product, such as food, therein. The closure **14** is configured to mount selectively to the container **12** to block removal of the product from the container interior **16**. The closure **14** is initially installed on the container **12** in an unopened, sealed state using a seal layer **44** of the closure **14**. The seal layer **44** may be co-extruded with one or more other layers (i.e. a first layer **40**A and one or more additional layers **40**N) to establish a multi-layer sheet that is used to form the closure **14** during a forming and assembly process **100** as shown in FIG. **6**. Later in the forming and assembly process **100**, the seal layer **44** is heated after the closure **14** is attached to the container **12** with the product stored in the container interior **16** to form a seal ring **44′** between the closure **14** and the container **12** to block effluents from entering the container interior **16** until a consumer peels the closure **14** from the container **12** to open the package **10** for the first time and establish a first-use, opened state. The closure **14** and/or the container **12** may include retaining structures that allow the closure **14** to be reinstalled on the container **12** after the package **10** has been opened by the consumer in an unsealed, closed state while the seal layer **44** remains attached to the closure so that the closure **14** and the seal layer **44** are not discarded separately from one another and can be recycled with one another

[0036] The container **12** includes a container brim **18**, a container floor **20** spaced apart from the container brim **18**, and a container sidewall **22** extending between and interconnecting the container brim **18** and the container floor **20** as shown in FIGS. **1-3**. The container interior **16** is located radially inward from the container brim **18** and the container sidewall **22** and above the container floor **20**. The container brim **18** is spaced radially from and surrounding a central axis **24** of the container **12**. The container brim **18** is coupled to an upper end **26** of the container sidewall **22** to extend radially outward away from the central axis **24** to interface with portions of the closure **14** and form the seal ring **44′** during the process **100**.

[0037] The closure **14** is configured to couple to the container brim **18** to close selectively an open mouth **13** formed in the container and defined by the container brim **18** as shown in FIGS. **1** and **2**. The closure **14** includes a closure top wall **30**, a closure sidewall **32** coupled to a radially outer edge **34** of the closure top wall **30** and a container-sealing flange **36** coupled to an upper end of the closure sidewall **32**. The closure top wall **30** extends substantially perpendicular to the central axis **24** of the container **12** when the closure **14** is installed thereon and provides a top surface **38** that can be used for labeling and/or decorations. In some embodiments, one or more of the sheets forming the closure can be pre-labeled or pre-decorated and applied via in-mold-labeling as the closure **14** is thermoformed. The closure sidewall **32** extends downwardly from the closure top wall **30** and is configured to engage a radially inner surface of the container sidewall **22** to mount the closure **14** to the container **12**. The container-sealing flange **36** extends radially outward away from the closure sidewall **32** to mount with the container brim **18** of the container **12** and provide the seal ring **44′** between the closure **14** and the container brim **18**.

[0038] The container-sealing flange **36** includes a support layer **40** coupled to an upper end **42** of the closure sidewall **32** and a seal layer **44** coupled to a lower surface **46** of the support layer **40** as shown in FIGS. **3** and **5**. The support layer **40** is formed integrally with and of the same material as the closure top wall **30** and the closure sidewall **32**. The seal layer **44** includes a different polymeric material from the rest of the closure **14** and is bonded to the lower surface **46** of the support layer **40** during the process **100** to form the seal ring **44′**. The seal layer **44** makes up about 10% of the closure **14** (i.e. 10% of the weight and/or thickness).

[0039] The process **100** of forming and assembling the package **10** is described in FIG. **6**. The process begins with a steps **102**, **104** of forming the container **12** and closure **14**. The container **12** and closure **14** are illustratively both formed from one or more polymeric materials, such as polypropylene or polyethylene. In some embodiments, the support layer **40** of the closure **14** consists of polypropylene and/or polyethylene. In some embodiments, the support layer **40** of the closure **14** include about 90% polypropylene and about 10% polyethylene. In some embodiments, the support layer **40** of the closure **14** includes up to 10% polyethylene. In one example, the container **12** is formed by thermoforming a sheet of the polymeric material. In other examples, the injection molding, by blow molding, or any other suitable forming method.

[0040] The closure **14** is formed during step **104** by at least two separate polymeric sheets or layers. In some embodiments, two polymeric layers are co-extruded together and then thermoformed as shown in FIG. **6**. In other embodiments, the polymeric layers are extruded separately and then laminated together and then thermoformed after or during the lamination step.

[0041] The step **104** of forming the closure **14** includes a sub-step **106** of extruding a first polymeric material and a sub-step **108** of extruding a second polymeric material. The first and second material may be in the form of a molten layers if being co-extruded or in the form of sheets if being laminated. The step **104** further includes a sub-step **110** of contacting or combining the first material with the second material to form a multi-layer sheet including at least the first polymeric material and the second polymeric material. If the materials are co-extruded, the first polymeric material can be contacted with the second polymeric material so that the materials cure and bond to one another. If the materials are laminated to one another, an intervening bond layer can be applied between the two polymeric sheets. As such, any number of layers can be combined together with the first polymeric material and the second polymeric material and then laminated to one another. Some suitable bond layers include a compatible layer of polypropylene or polyethylene, or an adhesive such as epoxy.

[0042] The first polymeric material is configured to provide a main and/or structural layer of the closure **14** and has a first melt-initiation temperature. The second polymeric material is configured to provide a bond layer and has a second melt-initiation temperature less than the first melt-initiation temperature of the main layer. The step **104** further includes a sub-step **112** of thermoforming the multi-layer sheet to form one or more closure preforms having the main layer and the bond layer and a web interconnecting each of the closure preforms. The step **104** may further include a sub-step **114** of trimming the closure preforms from the web to provide one or more closures **14** each having the main layer and the bond layer.

[0043] The process **100** may further include a step **116** of filling the container **12** with the product and a step of mounting the closure **14** to the container **12** to close the open mouth **13** with the product inside the container interior **16**. The process **100** further includes a step **120** of sealing the bond layer of the closure **12** to the container brim **18** by heating the container-sealing flange **36** of the closure **14** to the second melt-initiation temperature. The step **120** can include heat sealing with a heating device or source, sonic welding, ultrasonic welding or any other suitable manner for heating the bond layer to seal the bond layer to the container brim **18**. In some embodiments, arrangement of layers allows an external heating device to contact the support layer and heat the bond layer though the support layer at a temperature that is less than the melt point of the support layer but at or above the melt point of the bond layer. This allows the bond layer to melt and provide a heat seal between the support layer and the container brim **18** without leaving residue on the heating device because of the intervening support layer **40**. Sonic or ultra sonic welding can provide the same benefit due to the lack of an external heating device.

[0044] In some embodiments, the closure preforms remain connected to the web during the step **118** of mounting to container(s) **12**, and the step **114** of trimming comes after the step **116** of filling the container **12** and after or during the step **120** of sealing the bond layer to the container. The process **100** can also include a step of decorating the top wall **30** of the closure **14**, the closure preforms, the multi-layer sheet, or one or both of the first and second sheet.

[0045] The container sealing flange **36** further includes a pull tab **48** that extends radially outward away from the central axis **24** beyond the container brim **18** when the closure **14** is installed on the container **12** as shown in FIGS. **1-5**. The closure **14** can be removed from the container **12** by grasping the pull tab **48** and pulling upwardly away from the container brim **18** to release the seal ring **44′** from the brim **18** and peel the closure **14** away from the container **12**. The pull tab **48** is formed with a plurality of ribs **49** to block against deformation of the pull tab **48**.

[0046] The closure sidewall **32** is structured to allow the closure **14** to be reinstalled and retained on the container **12** without the seal ring **44′** as shown in FIGS. **3** and **5**. The closure sidewall **32** includes an inner sidewall **50** coupled to the radially outer edge **34** of the closure top wall **30**, an outer sidewall **52** coupled to the container-sealing flange **36**, and a sidewall link **54** interconnecting lower ends **56**, **58** of the inner sidewall **50**, and the outer sidewall **52**, respectively. The inner sidewall **50** is spaced apart radially from the container **12**. The outer sidewall **52** is configured to engage an interior surface **60** of the container sidewall **22** and/or the container brim **18**. The sidewall ink **54** forms the lowermost portion of the closure **14** and resides within the container interior **16** when the closure **14** is installed.

[0047] The inner sidewall **50** and spaced radially outward from the inner sidewall to provide an annular cavity **64** radially between the inner and outer sidewalls **50**, **52** and above the sidewall link **54** as shown in FIG. **5**. The container-sealing flange **36** is located above the top wall **30** and the annular cavity **64** such that the annular cavity is located at least partially below the container brim **18** when the closure **14** is installed. Illustratively, at least a portion of the container brim **18** is positioned above the top wall **30** of the closure **14** when the closure is installed.

[0048] The inner sidewall **50** is formed to include a first set of reinforcement ribs **66** spaced circumferentially apart from one another around the closure sidewall **32** as shown in FIGS. **4** and **5**. The outer sidewall **52** is formed to include a second set of reinforcement ribs **68** spaced circumferentially apart from one another around the closure sidewall **32**. Each reinforcement rib included in the first set of reinforcement ribs **66** has a first circumferential length **90** and each reinforcement rib included in the second set of reinforcement ribs **68** has a second circumferential length **92** greater than the first circumferential length **90**. The first and second sets of reinforcement ribs **66**, **68** are configured to block bending of the closure sidewall **32** as the closure **14** is installed and removed on the container **12**.

[0049] The first set of reinforcement ribs **66** extend radially outward away from the central axis **24** and into the annular cavity **64** as shown in FIGS. **4** and **5**. The second set of reinforcement ribs **68** extend radially inward toward the central axis **24** and into the annular cavity **64**. The first and second sets of reinforcement ribs **66**, **68** provide a stack shoulder for a second closure with a substantially similar closure sidewall. While stacked with the closure **14**, the closure sidewall of the second closure rests on the first and second reinforcement ribs **66**, **68**. The reinforcement ribs **66**, **68** are located such that an upper end of each reinforcement rib **66**, **68** is spaced apart from the container-sealing flange **36** by a distance sufficient to maintain spacing between the closure **14** and a seal ring also included in the second closure to block bonding between each closure. The inner sidewall **50** further includes a pair of diametrically opposed rotation blockers **67** that block rotation of the second closure relative to the closure **14** when stacked.

[0050] The container sidewall **22** is formed to include a first annular retainer ring **70** and the outer sidewall **52** of the closure sidewall **32** is formed to include a second annular retainer ring **72** as shown in FIGS. **3** and **5**. The first and second annular retainer rings **70**, **72** are structured similarly to provide an interlocking interface between the outer sidewall **52** and the container sidewall **22** when the closure **14** is installed. The interlocking interface is configured to retain the closure **14** on the container **12** and/or provide a seal between the closure **14** and the container separate from the seal ring **44′**. The second annular retainer ring **72** has a greater diameter than a portion of the first annular retainer ring **70** so that the closure **14** interlocks with the container **12** when it is installed. Installing and removing the closure **14** causes slight deformations of the container **12** as the second annular retainer ring **72** passes the first annular retainer ring **70**.

[0051] The first annular retainer ring **70** projects inwardly toward the central axis **24**. The second annular retainer ring **72** includes an outwardly-extending band **78** and an inwardly-extending band **80**. An outer surface of the inwardly-extending band **80** of the second annular retainer ring **72** is configured to engage an inner surface **60** of the first annular retainer ring **70** to block separation of the closure **14** from the container **12**. An upper end of the first annular retainer ring **70** is coupled directly to the container-sealing flange **36**. The second annular retainer ring **72** is located vertically between the second set of reinforcement ribs **68** and the container-sealing flange **36**. The container **12** further includes a stacking shoulder **74** configured to engage with the first annular retainer ring **70** of another container when stacked.

[0052] The closure **14** of the illustrative embodiment is configured to replace a liner and cap solution included with other comparative packages used to store food products, such as dairy products (i.e., yogurt, cheese, etc.). The closure **14** may have a dark or black color to block UV light from penetrating therethrough, which could spoil such food products. The package **10** may lack materials that are not suitable for food-grade packages, such as certain non-food grade adhesives. Accordingly, the present disclosure provides means for sealing a container **10** with only a closure **14** and without any liners (i.e. a pealable foil or film liner). In some embodiments, the package **10** consists of the container **12** and the closure **14**.

[0053] The closure **14** of the illustrative embodiment may be thermoformed from a co-extruded multilayer sheet including an outer layer and a seal layer. In some embodiments the seal layer includes a pealable resin manufactured by DOW® titled APPEAL™ 35D220. Unexpectedly it was found that such resins provided a suitable seal layer for the thermoformed closure **14** as well. Prior to conceiving of the package **10** disclosed herein, such pealable resins were not expected to maintain suitable properties to function as a seal layer **44** after undergoing a thermoforming process.

[0054] Another embodiment of a package **210** in accordance with the present disclosure, is shown in FIGS. **8-11**. The package **210** is similar to package **10** and includes a container **212** and a closure **214**. The container **212** is formed to include a container interior **216** configured to store a product, such as food, therein. The closure **214** is configured to mount selectively to the container **212** to block removal of the product from the container interior **216**. The closure **214** is initially installed on the container **212** in an unopened, sealed state using a seal layer **244** of the closure **214**. The seal layer **244** may be co-extruded with one or more other layers to establish a multi-layer sheet that is used to form the closure **214** during the forming and assembly process **100** as shown in FIG. **6**. Later in the forming and assembly process **100**, the seal layer **244** is heated after the closure **214** is attached to the container **212** with the product stored in the container interior **216** to form a seal ring **244′** between the closure **214** and the container **212** to block effluents from entering the container interior **216** until a consumer peels the closure **214** from the container **212** to open the package **210** for the first time. The closure **214** and the container **212** may each include retaining structures that allow the closure **214** to be reinstalled on the container **212** after the package **210** has been opened by the consumer.

[0055] The container **212** includes a container brim **218**, a container floor **220** spaced apart from the container brim **218**, and a container sidewall **222** extending between and interconnecting the container brim **18** and the container floor **220** as shown in FIGS. **8-10**. The container interior **216** is located radially inward from the container brim **18** and the container sidewall **222** and above the container floor **220**. The container brim **218** is spaced radially from and surrounding a central axis **224** of the container **212**. The container brim **18** is coupled to an upper end **226** of the container sidewall **222** to extend radially outward away from the central axis **224** to interface with portions of the closure **214** and form the seal ring **244′** during the process **100**.

[0056] The closure **214** is configured to couple to the container brim **218** to close selectively an open mouth **213** formed in the container and defined by the container brim **18** as shown in FIGS. **8** and **9**. The closure **214** includes a closure top wall **230**, a closure sidewall **232** coupled to the closure top wall **230**, and a container-sealing flange **236** coupled to an upper end of the closure sidewall **232**. The closure top wall **230** extends substantially perpendicular to the central axis **224** of the container **212** when the closure **214** is installed thereon and provides a top surface **238** that can be used for labeling and/or decorations. In some embodiments, one or more of the sheets or layers forming the closure **214** can be pre-labeled or pre-decorated and applied via in-mold-labeling as the closure **214** is thermoformed. The closure sidewall **232** extends downwardly from the closure top wall **230** and is configured to engage the brim **218** to mount the closure **214** to the container **212**. The container-sealing flange **236** extends between and interconnects the closure sidewall **32** and the top wall **230** to mount with an upper surface of the container brim **218** of the container **212** and provide the seal ring **244′** between the closure **214** and the container brim **218**.

[0057] The container-sealing flange **236** includes a support layer **240** coupled to an upper end **242** of the closure sidewall **232** and a seal layer **244** coupled to a lower surface **246** of the support layer **240** as shown in FIGS. **10** and **12**. The support layer **240** is formed integrally with and of the same material as the closure top wall **230** and the closure sidewall **232**. The seal layer **244** includes a different polymeric material from the rest of the closure **214** and is bonded to the lower surface **246** of the support layer **240** during the process **100** to form the seal ring **244′**. The seal layer **244** makes up about 10% of the closure **14** (i.e. 10% of the weight and/or thickness).

[0058] The closure **214** further includes a pull tab **248** that extends radially outward away from the central axis **224** beyond the container brim **218** when the closure **214** is installed on the container **212** as shown in FIGS. **8-10**. The closure **214** can be removed from the container **212** by grasping the pull tab **248** and pulling upwardly away from the container brim **218** to release the seal ring **244′** from the brim **218** and peel the closure **214** away from the container **212**. The pull tab **248** is formed with a plurality of ribs **249** to block against deformation of the pull tab **248**.

[0059] The closure sidewall **232** is structured to allow the closure **214** to be reinstalled and retained on the container **212** without the seal ring **244′** after the closure is opened for the first time as shown in FIGS. **9** and **10**. The closure sidewall **232** includes a sidewall body **263**, an annular sidewall ring **265**, and a plurality of reinforcement ribs **266** spaced circumferentially apart from one another around the sidewall body **263** as shown in FIGS. **10** and **11**. The annular sidewall ring **265** projects inwardly toward the central axis **224**. The plurality of reinforcement ribs **266** are coupled to the annular sidewall ring **265** and project inwardly from the annular sidewall ring **265**. The plurality of reinforcement ribs **266** are configured to block bending of the closure sidewall **232** as the closure **214** is installed and removed on the container **212**.

[0060] The plurality of reinforcement ribs **266** are also configured to provide a secondary retention unit configured to retain the closure **214** on the container **12** after the closure **214** has been peeled away from the brim **218** for the first time. When the closure **214** is installed on the container **212**, the plurality of reinforcement ribs **266** are located below a distal end **219** of the brim **218** as shown in FIG. **10**. An inner end of each rib **266** is spaced a smaller distance from the central axis **224** compared to the distal end **219** to such that the ribs **266** interlock with the brim **218** and block removal of the closure **214** from the container **212**. Pulling upwardly on the peel tab **248** causes slight deformations of the closure **214** sufficient to separate the ribs **266** surrounding the peel tab **248** from the brim **218** so that the closure can be separated from the container **212**. The distal end **219** of the brim **218** is configured to engage a portion of the sidewall body **263** above the plurality od ribs **266** to provide a secondary sealing interface therebetween after the closure **214** is opened for the first time.

[0061] The plurality of reinforcement ribs **266** also provide a stack shoulder for a second closure with a substantially similar closure sidewall. The sidewall **232** further includes a tab stiffener **267** that projects outwardly away from the central axis **224** toward the peel tab **248** to reinforce a junction between the peel tab **248** and the closure sidewall **232**. The tab stiffener **267** also blocks rotation of the closure **214** relative to other similar closures when stacked with the other similar closures. The container **212** further includes a stacking shoulder **274** configured to engage with the brim **218** of another container when stacked.

Claims

1. A package comprising: a container including a container brim, a container floor spaced apart from the container brim, and a container sidewall extending between and interconnecting the container brim and the container floor to define a container interior radially inward from the container brim and the container sidewall and above the container floor, the container brim spaced radially from and surrounding a central axis of the container and coupled to an upper end of the container sidewall to extend radially outward away from the central axis, and a closure configured to couple to the container brim to close selectively an open mouth of the container defined by the container brim, the closure including a closure top wall, a closure sidewall coupled to the closure top wall and extending downwardly from the closure top wall to engage the container to mount the closure to the container, and a container-sealing flange coupled to an upper end of the closure sidewall and configured to mount with the container brim of the container and close the open mouth, wherein the container-sealing flange includes a support layer coupled to the closure sidewall and a seal ring bonded to a lower surface of the support layer and configured to establish: (i) a sealed, unopened state of the package in which the seal layer is sealed with an upper surface of the container brim to block effluents from entering the container interior, (ii) a first-use opened state in which the closure is separated from the container and the seal layer remains attached to the support layer, and (iii) an unsealed, closed state in which the closure is reinstalled on the container and the seal layer remains attached to the support layer so that the closure and the seal layer are not discarded separately from one another and can be recycled with one another.
2. The package of claim 1, wherein the closure sidewall is coupled to a radially outer edge of the closure top wall and a radially inner edge of the container sealing flange.
3. The package of claim 2, wherein the closure sidewall includes an inner sidewall coupled to the radially outer edge of the closure top wall, an outer sidewall coupled to the support layer and spaced radially outward from the inner sidewall, and a sidewall link interconnecting a lower end of the inner sidewall and the outer sidewall.
4. The package of claim 3, wherein the inner sidewall is formed to include a first set of reinforcement ribs spaced circumferentially apart from one another around the closure sidewall.
5. The package of claim 4, wherein the outer sidewall is formed to include a second set of reinforcement ribs spaced circumferentially apart from one another around the closure sidewall.
6. The package of claim 5, wherein each reinforcement rib included in the first set of reinforcement ribs has a first circumferential length and each reinforcement rib included in the second set of reinforcement ribs has a second circumferential length greater than the first circumferential length.
7. The package of claim 5, wherein the first and second sets of reinforcement ribs provide a stack shoulder for a second closure and an upper end of each reinforcement rib is spaced apart from the container-sealing flange by a distance sufficient to maintain spacing between the closure and a seal ring included in the second closure.
8. The package of claim 4, wherein the container sidewall is formed to include a first annular retainer ring and the outer sidewall is formed to include a second annular retainer ring configured to engage the second annular retainer ring when the closure is installed on the container to block removal of the closure from the container, and the second annular retainer ring is located vertically between the second set of reinforcement ribs and the container-sealing flange.
9. The package of claim 1, wherein the seal ring provides a heat seal configured to bond to the container brim in the sealed, unopened state to block passage of effluents into the container interior in the sealed, unopened state of the container, and the package lacks an separate foil or film layer between the closure and the container.
10. A closure configured to couple to a container to close selectively an open mouth of the container, the closure comprising: a closure top wall, a closure sidewall coupled the closure top

wall and extending downwardly from the closure top wall to engage the container to mount the closure to the container, and a container-sealing flange coupled to an upper end of the closure sidewall and extending radially outward away from the closure top wall to mount with the container brim of the container and close the open mouth, wherein the container-sealing flange includes a support layer coupled to the closure sidewall and a seal ring coupled to a lower surface of the support layer to engage with an upper surface of the container brim and provide a sealing layer therebetween.

11. The closure of claim 10, wherein the closure sidewall is coupled to a radially outer edge of the closure top wall and a radially inner edge of the container sealing flange, and the closure sidewall includes an inner sidewall coupled to the radially outer edge of the closure top wall, an outer sidewall coupled to the support layer and spaced radially outward from the inner sidewall, and a sidewall link interconnecting a lower end of the inner sidewall and the outer sidewall.

12. The closure of claim 11, wherein the inner sidewall is formed to include a first set of reinforcement ribs spaced circumferentially apart from one another around the closure sidewall, and the outer sidewall is formed to include a second set of reinforcement ribs spaced circumferentially apart from one another around the closure sidewall.

13. The closure of claim 12, wherein each reinforcement rib included in the first set of reinforcement ribs has a first circumferential length and each reinforcement rib included in the second set of reinforcement ribs has a second circumferential length greater than the first circumferential length.

14. The closure of claim 11, wherein the container sidewall is formed to include a first annular retainer ring and the outer sidewall is formed to include a second annular retainer ring configured to engage the second annular retainer ring when the closure is installed on the container to block removal of the closure from the container.

15. The closure of claim 10, wherein the container-sealing flange extends between and interconnects the closure top wall and an upper end of the closure sidewall, and wherein the closure sidewall includes a plurality of ribs located below a distal end of the container brim when the closure is installed to interlock with the container brim and provide a secondary retainer unit after the closure is peeled from the container for the first time.

16. The closure of claim 10, wherein the seal ring is heated to bond to a container brim in a sealed, unopened state to block passage of effluents into a container interior in the unopened state of the container.

17. A method of forming and installing a closure on a container to close selectively an open mouth of the container, the method comprising extruding a first polymer material having a first melt strength, extruding a second polymer material having a second melt strength less than the first melt strength, combining the first polymeric material with the second polymeric material to provide a multi-layer sheet, thermoforming the multi-layer sheet to provide the closure, the first polymer material forms a closure top wall, a closure sidewall coupled to a radially outer edge of the closure top wall and extending downwardly from the closure top wall to engage the container to mount the closure to the container, and a support layer coupled to an upper end of the closure sidewall and extending radially away from the closure sidewall, and the second polymer material forms a seal layer coupled to a lower surface of the support layer, mounting the closure on the container such that the seal layer engages with an upper surface of a container brim included in the container, and heating the seal layer to cause the seal layer to bond with the container brim and provide a seal ring therebetween.

18. The method of claim 17, wherein the first and second polymeric materials are co-extruded together and contact one another in a molten or partially-molten state.

19. The method of claim 17, wherein the first and second polymeric materials are extruded separately from one another and are laminated together before or during the step of thermoforming.

20. The method of claim 17, wherein heating the seal layer includes ultrasonically welding the seal layer to the container brim.